

Scientific Computation

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Lecture 32 - Numerical Differentiation

Approximating the First Derivative: The Forward Difference

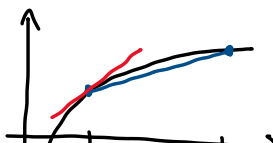
When learning about truncation error at the start of this course, we approximated the first derivative of a function $f(x)$ as

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

Using the Taylor series, we found that the error in this approximation was:

$$\left| f'(x) - \frac{f(x+h) - f(x)}{h} \right| = \left| \frac{-f''(\xi) \cdot h}{2} \right|, \quad \xi \in [x, x+h]$$

Forward difference method
(first order accurate: $O(h)$)



$$\frac{f(x) - f(x-h)}{h}$$

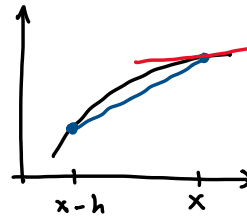
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The Backward Difference

We can similarly introduce the **backward difference** as

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

Let's repeat our Taylor series-based analysis of the error:



$$\begin{aligned} f(x-h) &= f(x) + f'(x)(x-h-x) + \frac{1}{2}f''(\xi)(x-h-x)^2, \quad \xi \in [x-h, x] \\ &= f(x) - h \cdot f'(x) + \frac{h^2}{2}f''(\xi) \end{aligned}$$

$$\Rightarrow \frac{f(x) - f(x-h)}{h} = f'(x) + \frac{h}{2}f''(\xi) \rightarrow \text{1st order as well O(h)}$$

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Higher Order Approximations of the First Derivative

$$x_{i+1} \triangleq x_i + h$$

$$x_{i-1} \triangleq x_i - h$$

What if we want an approximation whose error shrinks faster as $h \rightarrow 0$? Just like there are higher order Runge-Kutta methods, we can use more terms in a Taylor series expansion of f to get a higher-order method that uses more points around x_i to compute $f'(x_i)$. For example, let's write two Taylor expansions at x_{i+1} and x_{i-1} :

$$f(x_{i+1}) = f(x_i) + f'(x_i) \cdot h + \frac{1}{2}f''(x_i)h^2 + \frac{h^3}{6}f'''(\xi_+)$$

$$f(x_{i-1}) = f(x_i) - f'(x_i) \cdot h + \frac{1}{2}f''(x_i)h^2 - \frac{h^3}{6}f'''(\xi_-)$$

$$\text{where } \xi_+ \in [x_i, x_{i+1}], \xi_- \in [x_{i-1}, x_i]$$

$$\Rightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} + \frac{h^2}{12}[f'''(\xi_+) + f'''(\xi_-)]$$

error term

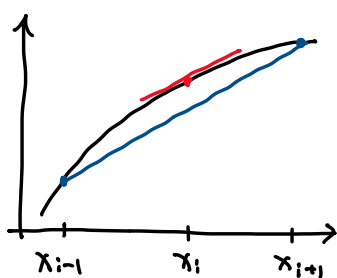
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The Central Difference Method

This is called the **central difference formula**:

$$f'(x_i) \approx \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}.$$

In Q.7a of Assignment 1, we proved that this is a **second-order accurate** method:



$$\left| f'(x_i) - \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} \right|$$

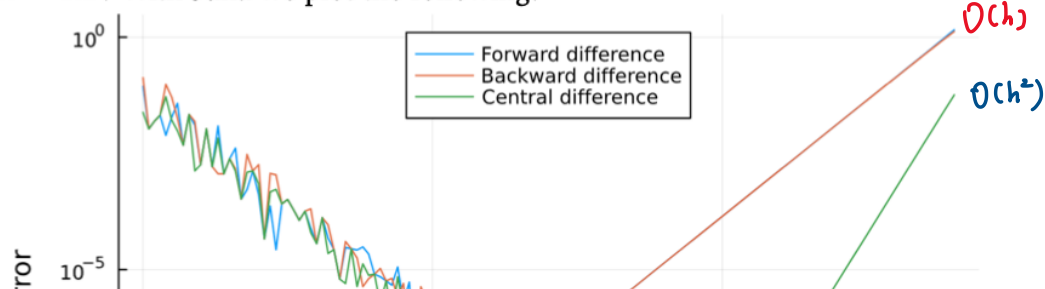
$$= \frac{h^2}{12} \left| f'''(\xi_+) + f'''(\xi_-) \right|$$

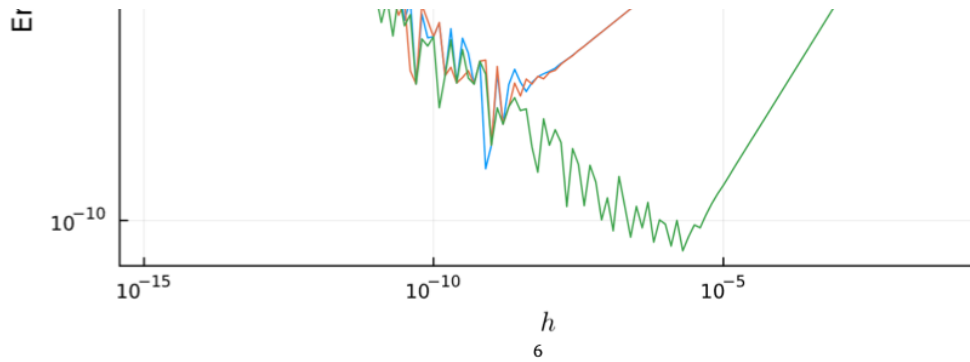
Apply the IVT or triangle inequality to simplify further

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Comparing Finite Difference Formulae

Example 1: How accurate is each approximation of f' for $f(x) = \sin(x)e^x$ at $x = 0.5$? With Julia we plot the following:





Approximating the Second Derivative

What if we need to approximate the second derivative? We have the **second-order central difference** formula:

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}.$$

Just as before, we get this expression from Taylor series expansions:

$$\begin{aligned} f(x_{i+1}) &= f(x_i) + f'(x_i)h + \frac{h^2}{2}f''(x_i) + \frac{h^3}{6}f'''(x_i) + \frac{h^4}{24}f^{(4)}(\xi_+) \\ f(x_{i-1}) &= f(x_i) - f'(x_i)h + \frac{h^2}{2}f''(x_i) - \frac{h^3}{6}f'''(x_i) + \frac{h^4}{24}f^{(4)}(\xi_-) \end{aligned}$$

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Error in the Second-Order Central Difference Formula

Adding these expressions together and rearranging gives us

$$f(x_{i+1}) + f(x_{i-1}) = 2f(x_i) + f''(x_i)h^2 + \underbrace{\frac{1}{24}[f^{(4)}(\eta_1) + f^{(4)}(\eta_2)]h^4}_{\text{error term}},$$

where terms with an odd power of h have cancelled out due to the symmetry in $(x_{i+1} - x_i) = h = -(x_i - x_{i-1})$.

$$\Rightarrow \left| f''(x_i) - \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{2h} \right| \leq \frac{h^2}{12} \max |f'''(\xi)|, \xi \in [x_{i-1}, x_{i+1}]$$

↓

Applying the IVT or triangle, ineq.

我们知道用差分计算一阶导数是 $O(h)$ 的，有没有办法能提高它的阶数？

于是我们引入 **central difference method**，核心思想是用对称的两个点， $x_{i+1} = x_i + h, x_{i-1} = x_i - h$

来一起逼近 $f'(x_i)$ 。分别在这两个对称点展开泰勒级数，再让 $f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h +$

$\frac{1}{6}h^3[f'''(\xi_+) + f'''(\xi_-)]$ ，于是得到 $f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} + \frac{h^2}{12}[f'''(\xi_+) + f'''(\xi_-)]$